

TrustGraspNet: Trust-Aware IMU-Only Grasp Assessment for Soft Spiral Robots under Sparse Sensing

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Abstract—Soft spiral robots can grasp fragile and irregular objects with remarkable adaptability, but reliable grasp assessment remains difficult because grasp state is expressed through distributed body deformation rather than a small set of rigid contacts. We present TrustGraspNet, a compact trust-aware multi-task sequence model that predicts grasp success, grasp quality, intervention action, object family, object size, and per-IMU reliability from onboard proprioception alone. The method is intentionally framed as a simulation-first learning study: the central question is whether sparse body dynamics contain enough information for decision-level grasp assessment before introducing heavier sensing or custom tactile hardware. The proposed model uses eight D-optimal IMUs and two tendon-displacement commands, combines shared per-sensor encoding with validity-aware fusion and a lightweight Conv1D-GRU temporal backbone, and is trained under explicit sensor corruption with per-sensor trust supervision. Across 1,000 MuJoCo grasp trials and 12,330 labeled windows, a five-seed evaluation on a leakage-free trial-level split shows strong performance. After aggregating overlapping windows within each trial, the 8-IMU displacement-only student reaches 0.9940 ± 0.0009 AUROC for grasp success, 0.498 ± 0.011 exact-7 family accuracy, 0.1496 ± 0.0025 quality RMSE, and 0.00354 ± 0.00016 size RMSE, while the window-level action head reaches 0.9481 ± 0.0024 accuracy. Under fully corrupted evaluation, success remains stable at 0.9946 ± 0.0015 AUROC and action accuracy at 0.9432 ± 0.0024 , whereas trust RMSE rises from 0.0477 ± 0.0027 to 0.2093 ± 0.0003 , correctly reflecting degraded sensing reliability. Ablations show that corruption-aware training is the dominant source of robust task performance, explicit trust supervision is essential for meaningful trust estimates, and the D-optimal 8-IMU subset preserves the information available in a denser 24-IMU layout. These results position sparse IMU-only grasp assessment as a credible contribution at the intersection of soft robotics, robust trustworthy learning, and deployment-oriented TinyML.

Index Terms—soft robotics, grasp assessment, proprioception, inertial measurement unit, trust-aware learning, TinyML

I. INTRODUCTION

Soft spiral robots are attractive manipulators because their compliant bodies can wrap around fragile, irregular, and uncertain objects without relying on the tightly localized contacts typical of rigid grippers [1]–[4]. The same compliance that gives soft robots their versatility also makes feedback difficult. In a rigid hand, grasp state can often be summarized by a small set of contact or force variables. In a soft spiral body, by contrast, grasp state is distributed across deformation, delayed load transfer, partial slip, and enclosure quality. A practically useful controller therefore needs more than a binary success

label: it needs an early estimate of whether the grasp is likely to succeed, whether the sensed deformation is trustworthy, and whether the robot should continue, tighten, retry, or abort.

Most existing grasp-assessment pipelines obtain this information from external perception or rich contact sensing. Vision, tactile arrays, and multimodal sensing have produced strong results in grasp prediction and manipulation, but they also introduce calibration burden, wiring complexity, packaging overhead, and additional computation [5], [6], [8]–[10]. These costs are especially problematic in compact soft platforms, where sensing hardware can quickly dominate the design itself. Proprioception is an appealing alternative because it is inherently onboard and naturally aligned with closed-loop control. Among practical proprioceptive modalities, IMUs are especially attractive: they are inexpensive, compact, and off-the-shelf. However, prior IMU-based work in soft robotics has focused mainly on shape estimation, state reconstruction, and feedback control rather than decision-level grasp assessment [11]–[14]. Moreover, IMU degradation—noise, drift, freezing, delay, dropout, or spoofing—is usually treated as a nuisance to be filtered away, not as a source of uncertainty that should itself be estimated.

This paper addresses that gap with **TrustGraspNet**, a trust-aware multi-task model for grasp assessment in a tendon-driven soft spiral robot. The work is deliberately scoped as a machine-learning paper rather than a hardware paper. Hardware constraints motivate the sensing budget, but all quantitative claims are established in simulation. That framing isolates the scientific question that matters most: *can sparse proprioception alone support reliable decision-level grasp assessment under realistic sensor corruption?* If the answer is negative, richer sensing is justified. If the answer is positive, sparse sensing becomes a principled design target rather than a compromise.

The proposed approach rests on three ideas. First, sensing is restricted to the regime that hardware can plausibly support: eight carefully selected IMUs and two tendon-displacement commands. Second, the architecture uses explicit inductive bias instead of generic sequence-model scale: shared per-sensor encoding, mask-aware fusion, and a lightweight Conv1D-GRU temporal backbone. Third, robustness is addressed directly by corrupting sensors during training and supervising a per-IMU trust head, so the model learns both to

remain task-relevant and to signal when the sensing substrate itself has become less reliable.

The contributions are therefore threefold. **(1)** We formulate IMU-only grasp assessment for a soft spiral robot as a trust-aware multi-task sequence-learning problem with outputs for success, quality, action, family, size, and per-sensor trust. **(2)** We combine D-optimal sensor placement with sparse validity-aware temporal fusion to preserve grasp-relevant information under an 8-IMU budget. **(3)** We provide five-seed evidence that sparse proprioception is sufficient for accurate grasp assessment, that corruption-aware training is the dominant source of task robustness, and that explicit trust supervision materially improves reliability estimation without requiring dense sensing or large models.

II. RELATED WORK

Sensing and feedback in soft grasping. Soft grippers exploit compliance to handle uncertain, delicate, and irregular objects, but this same compliance makes grasp state difficult to observe and regulate [5]–[7]. Recent systems have therefore integrated tactile, force, optical, and multimodal sensing to estimate slip, force, and object properties [9], [10]. These approaches are effective, yet they often rely on specialized fabrication, dense sensor routing, or heavier signal-processing stacks that are difficult to scale to lightweight soft manipulators.

Proprioception and distributed inertial sensing. A parallel literature studies proprioception through embedded strain sensing, deformable sensors, or inertial sensing distributed along the body [11], [12]. IMUs are particularly attractive because they are compact, inexpensive, and easy to integrate. Existing IMU-based methods have shown promise for soft-body reconstruction and feedback control [13], [14], but they do not directly solve the decision problem considered here: predicting from short windows of onboard dynamics whether a grasp will succeed and what corrective action should be taken.

Learning-based grasp assessment and edge deployment. Vision and tactile learning pipelines have demonstrated strong grasp-outcome prediction, but they depend on richer sensing and heavier compute [15]–[17]. In parallel, quantization and model compression have enabled compact edge models for robotics and embedded ML [18]–[20]. These two strands have rarely been combined in soft robotic grasp assessment, where the emphasis is often either rich perception or compliant hardware rather than sparse, deployable, corruption-aware inference.

Positioning of this work. The literature still lacks an end-to-end grasp-assessment pipeline that is simultaneously IMU-only, explicitly trust-aware, and designed for microcontroller-scale deployment in a soft spiral gripper. To the best of our knowledge, prior soft-robot IMU studies do not combine sparse sensor placement, window-level intervention prediction, explicit per-sensor reliability learning under realistic corruption, and a TinyML-oriented student model in a single framework. Our method is positioned precisely at that intersection.

III. METHODOLOGY

A. Problem Formulation

We consider a tendon-driven soft spiral robot instrumented with M IMUs. At time step t , sensor $m \in \{1, \dots, M\}$ provides angular velocity and linear acceleration,

$$\mathbf{x}_{t,m} = [\omega^x, \omega^y, \omega^z, a^x, a^y, a^z]_{t,m} \in \mathbb{R}^6, \quad (1)$$

and a binary validity variable $\mu_{t,m} \in \{0, 1\}$ indicates whether that stream is currently available. The actuator input is the commanded displacement of the two tendons,

$$\mathbf{u}_t = [u_t^{(1)}, u_t^{(2)}] \in \mathbb{R}^2. \quad (2)$$

Over a temporal window of length T , the model receives

$$\mathbf{X} \in \mathbb{R}^{T \times M \times 6}, \quad \mathbf{U} \in \mathbb{R}^{T \times 2}, \quad \boldsymbol{\mu} \in \{0, 1\}^{T \times M}, \quad (3)$$

and predicts grasp success probability $\hat{p}$, continuous grasp quality $\hat{q}$, intervention logits $\hat{\mathbf{y}}^{\text{act}}$, object-family logits $\hat{\mathbf{y}}^{\text{fam}}$, object size $\hat{r}$, and per-IMU trust $\hat{\tau}$. In the sparse deployment regime, $M = 8$ and $T = 80$, corresponding to a 0.40 s window at 200 Hz. The key challenge is early inference: the model must predict both task outcome and sensing reliability from a short partial observation of the robot’s own dynamics.

B. Dataset Generation and Supervision

Data are generated in MuJoCo using a 24-segment soft spiral robot and a seven-family object library: sphere, cylinder, capsule, box, flat, branched, and irregular. Each trial lasts 6 s and executes a nominal pack–unpack–reach–wrap–tighten–pull controller. The simulator logs distributed IMU motion, tendon-displacement commands, body positions, tip position, object state, and binary object–backbone contact. After the trial is complete, these trajectories are converted into supervision targets.

We define the continuous grasp-quality label as

$$q = \text{clip}\left(0.45r_{\text{att}} + 0.20r_{\text{con}} + 0.15 \text{clip}(\Delta z/0.05, 0, 1) + 0.10 \tanh(F_{\text{pull}}/4) + 0.10(1 - s_{\text{slip}}), 0, 1\right), \quad (4)$$

where r_{att} is the pull-phase attachment ratio, r_{con} is the whole-trial contact ratio, Δz is the maximum vertical lift, F_{pull} is a pull-force proxy, and s_{slip} penalizes separation growth during pulling. The binary success target is then defined conservatively as

$$y^{\text{succ}} = \mathbb{I}[r_{\text{att}} \geq 0.45 \wedge r_{\text{con}} \geq 0.10 \wedge q \geq 0.45]. \quad (5)$$

Each trajectory is segmented into overlapping windows of length 80 and stride 20. Every window inherits the trial-level success, quality, family, and size labels, and receives a phase-aware oracle action label from window-level contact, tip-object distance, and actuation statistics. The action space is *continue*, *tighten*, *retry*, and *abort*. This construction aligns training with intervention: the network must infer the eventual outcome from a partial temporal prefix rather than from a fully completed trial.

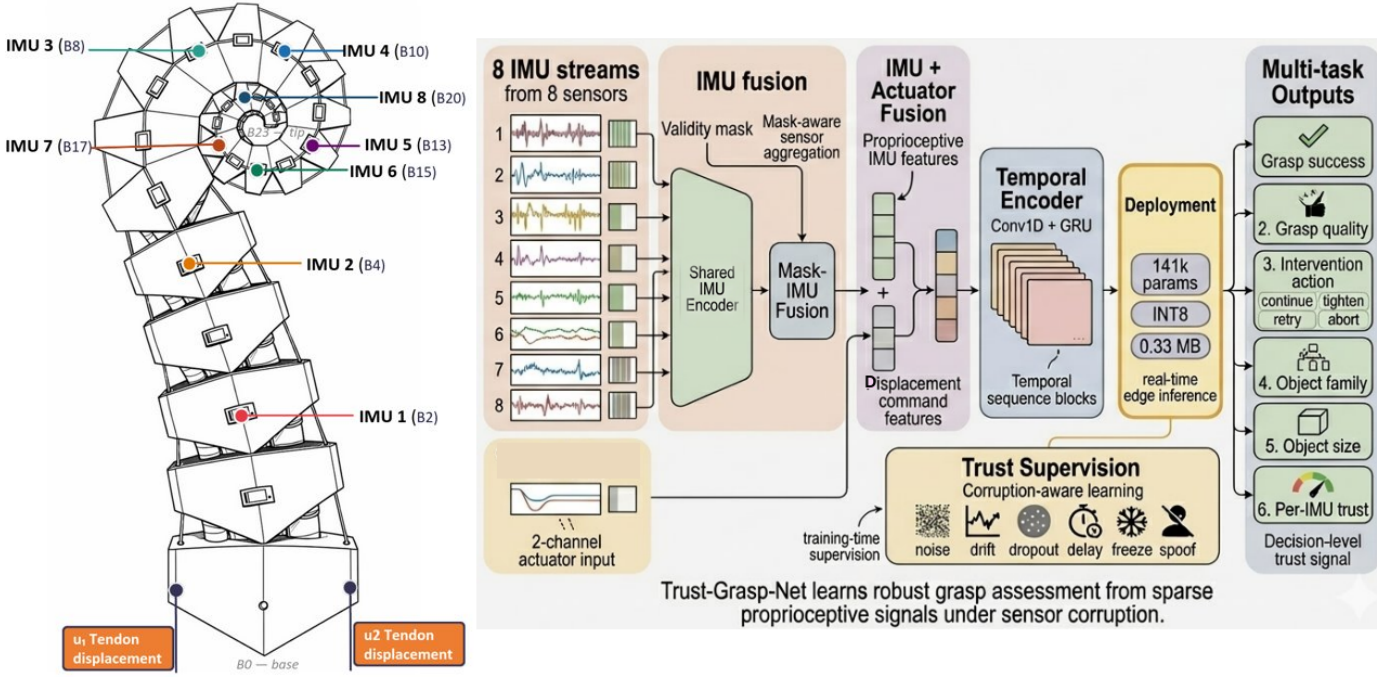


Fig. 1. Overview of TrustGraspNet. Eight D-optimal IMUs are mounted along the SpiRob backbone and fused with two tendon-displacement commands. The model uses shared IMU encoding, mask-aware fusion, a lightweight temporal encoder, and explicit trust supervision to predict success, quality, intervention action, object family, object size, and per-IMU trust from sparse proprioception alone.

C. Sparse IMU Placement and Trust-Aware Architecture

Because the physical platform cannot host dense sensing everywhere along the backbone, sensor placement is treated explicitly. Let $\Phi \in \mathbb{R}^{P \times k}$ denote the top- k principal deformation modes of the full 24-segment backbone. For a candidate subset S of segment locations, D-optimal selection maximizes the Fisher-information surrogate

$$J(S) = \log \det(\Phi_S^\top \Phi_S / \sigma^2), \quad (6)$$

subject to a spacing constraint that avoids redundant adjacent placements. The selected subset is $\{B2, B4, B8, B10, B13, B15, B17, B20\}$, concentrating sensors in base-proximal, lower-mid, mid-spine, and near-tip regions where curvature and load transfer vary most strongly during reaching, wrapping, and tightening.

Each IMU stream is encoded independently by a shared multilayer perceptron,

$$\phi_{t,m} = g_s(\mathbf{x}_{t,m}), \quad g_s : \mathbb{R}^6 \rightarrow \mathbb{R}^{24}, \quad (7)$$

which enforces a common local representation across sensors while keeping the parameter count small. A validity-aware fusion stage then aggregates only the streams marked available by the mask,

$$\bar{\phi}_t = \frac{\sum_{m=1}^M \mu_{t,m} \phi_{t,m}}{\max(1, \sum_{m=1}^M \mu_{t,m})}. \quad (8)$$

Actuator displacement is encoded separately as $\psi_t = g_a(\mathbf{u}_t)$ with $g_a : \mathbb{R}^2 \rightarrow \mathbb{R}^{32}$ and fused with the pooled IMU state as

$\mathbf{z}_t = g_f([\bar{\phi}_t; \psi_t])$, where $g_f : \mathbb{R}^{56} \rightarrow \mathbb{R}^{64}$. The fused sequence is processed by two lightweight Conv1D blocks followed by a GRU with hidden size 96,

$$\mathbf{h}_{1:T} = \mathcal{T}(\mathbf{z}_{1:T}), \quad \mathbf{g} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_t, \quad (9)$$

where $\mathbf{g}$ is the window-level representation used by the task heads. Separate lightweight heads predict success, quality, action, family, and size. Trust is predicted sensor-wise from time-aggregated per-sensor features,

$$\tilde{\phi}_m = \frac{\sum_{t=1}^T \mu_{t,m} \phi_{t,m}}{\max(1, \sum_{t=1}^T \mu_{t,m})}, \quad \hat{\tau}_m = h_{\text{trust}}(\tilde{\phi}_m). \quad (10)$$

This explicit trust head is important: reliability is a supervised output rather than an opaque by-product of a latent attention mechanism.

D. Corruption Model and Optimization

A central objective of the method is not merely to tolerate sensor degradation, but to expose it. During training, 50% of windows are corrupted online. For each selected window, 15–45% of IMUs are perturbed using additive noise, drift, clipping, delay, freezing, dropout, or spoofing, with severity $\alpha \sim \mathcal{U}[0.25, 1.0]$. Each corrupted sensor receives a supervisory trust target $\tau_m^* \in [0, 1]$ reflecting its residual expected utility. Hard dropout receives zero trust, spoofing receives near-zero trust, and milder corruptions decay smoothly with severity.

TABLE I
DATASET AND EVALUATION SUMMARY FOR THE STANDARD
LEAKAGE-FREE SPLIT USED IN THE MAIN EXPERIMENTS.

Quantity	Value
Simulation trials	1,000
Window length / stride	80 / 20 samples
IMU sampling rate	200 Hz
Total labeled windows	12,330
Train / val / test windows	8,834 / 824 / 1,526
Object families	7
Trial-level success rate	46.4%
Corruption training probability	50% of windows
Random seeds for final reporting	5

The network is trained with a weighted multi-task objective,

$$\mathcal{L} = \mathcal{L}_{\text{succ}} + \lambda_q \mathcal{L}_{\text{qual}} + \lambda_a \mathcal{L}_{\text{act}} + \lambda_f \mathcal{L}_{\text{fam}} + \lambda_r \mathcal{L}_{\text{size}} + \lambda_t \mathcal{L}_{\text{trust}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}}, \quad (11)$$

where success uses binary cross-entropy, action and family use cross-entropy, and quality, size, and trust use mean-squared error. The deployed student uses hidden widths 24/64/96 and contains 140,896 parameters. It is trained from a wider teacher (48/128/160 hidden widths) using a distillation loss on success, quality, and trust outputs. Optimization uses AdamW with learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 64, and 40 epochs. After INT8 quantization, the student occupies approximately 0.33 MB.

IV. EXPERIMENTAL PROTOCOL

The final dataset contains 1,000 trials and 12,330 windows after filtering. The standard leakage-free split is performed at the trial level, preventing leakage between overlapping windows. Unless stated otherwise, all quantitative results are averaged over five random seeds. Because windows from the same trial are strongly correlated, success, family, quality, and size are reported at the *trial level*: window predictions belonging to the same trial are aggregated before metric computation. Intervention action remains a *window-level* metric because its supervision is phase-local and assigned per temporal window. We report mean $\pm$ 95% confidence interval across seeds.

The empirical study is organized around four questions. **Q1:** Can sparse proprioception support accurate decision-level grasp assessment? **Q2:** Does corruption-aware training improve robust performance under sensor degradation? **Q3:** Is explicit trust supervision necessary to obtain meaningful trust estimates? **Q4:** Does the D-optimal 8-IMU configuration preserve the information available in denser sensing and richer actuator inputs? These questions motivate the main clean-vs-robust evaluation, the default/no-corruption/no-trust ablations, and the sensor-budget, actuation, and family-granularity studies reported next.

V. RESULTS

A. Main Performance under Clean and Corrupted Evaluation

Table II reports the main five-seed results for the 8-IMU displacement-only student. The model achieves near-perfect

TABLE II
MAIN RESULTS FOR THE 8-IMU DISPLACEMENT-ONLY STUDENT. SUCCESS, FAMILY, QUALITY, AND SIZE ARE REPORTED AT THE TRIAL LEVEL; ACTION AND TRUST ARE REPORTED AT THE WINDOW LEVEL. VALUES ARE MEAN $\pm$ 95% CONFIDENCE INTERVAL OVER FIVE SEEDS.

Metric	Clean	Robust
Success AUROC	0.9940 $\pm$ 0.0009	0.9946 $\pm$ 0.0015
Family accuracy	0.498 $\pm$ 0.011	0.492 $\pm$ 0.018
Quality RMSE $\downarrow$	0.1496 $\pm$ 0.0025	0.1521 $\pm$ 0.0025
Size RMSE $\downarrow$	0.00354 $\pm$ 0.00016	0.00353 $\pm$ 0.00017
Action accuracy	0.9481 $\pm$ 0.0024	0.9432 $\pm$ 0.0024
Trust RMSE $\downarrow$	0.0477 $\pm$ 0.0027	0.2093 $\pm$ 0.0003
Parameters	140,896	
INT8 size	0.33 MB	

TABLE III
ABLATION OF CORRUPTION-AWARE TRAINING AND TRUST SUPERVISION ON THE STANDARD TEST SPLIT. VALUES ARE MEAN $\pm$ 95% CONFIDENCE INTERVAL ACROSS FIVE SEEDS.

Setting	Rob. AUROC	Rob. Fam.	Rob. Act.	Rob. Qual.	Clean T.	Rob. T.
Default	0.9946 $\pm$ 0.0015	0.492 $\pm$ 0.018	0.9432 $\pm$ 0.0024	0.1521 $\pm$ 0.0025	0.0477 $\pm$ 0.0027	0.2093 $\pm$ 0.0003
No corr. train	0.9879 $\pm$ 0.0033	0.388 $\pm$ 0.039	0.7533 $\pm$ 0.0135	0.2009 $\pm$ 0.0073	0.0019 $\pm$ 0.0014	0.2767 $\pm$ 0.0036
No trust loss	0.9965 $\pm$ 0.0007	0.461 $\pm$ 0.024	0.9388 $\pm$ 0.0044	0.1520 $\pm$ 0.0017	0.518 $\pm$ 0.033	0.465 $\pm$ 0.027

trial-level separation of successful and failed grasps after aggregating overlapping windows within each trial. At the same time, the action head remains highly accurate at the window level, indicating that the short temporal windows contain sufficient information not only to assess final outcome but also to imitate a phase-aware intervention policy.

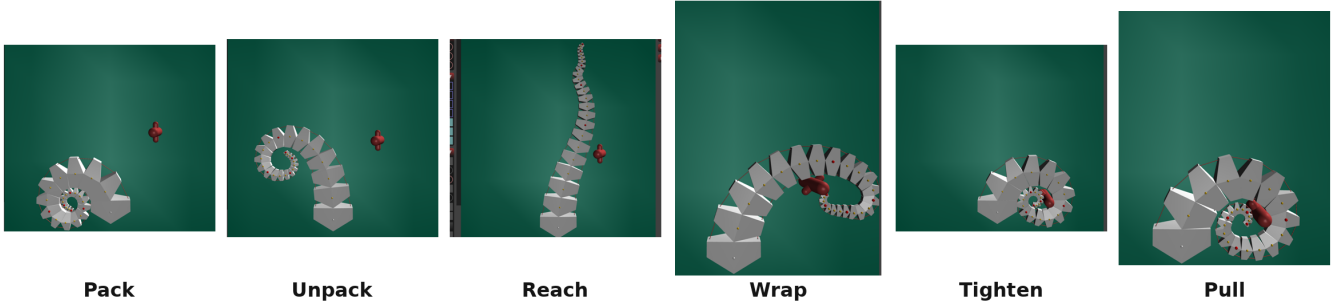
The most important observation is not the absolute value of success AUROC alone, but the stability of the entire task profile under corruption. Success, family, quality, size, and action shift only modestly between clean and robust evaluation, whereas trust RMSE rises sharply. This is the intended behavior: the model remains task-relevant while exposing that its sensor substrate has become more difficult to certify. In that sense, trust behaves as a decision-level reliability signal rather than as a redundant copy of the task heads.

B. What Drives Robustness? Corruption Training versus Trust Supervision

Table III and Fig. 3 isolate two distinct mechanisms in the method. The *no-corruption-training* ablation retains the same architecture but removes corruption-aware augmentation during optimization. The *no-trust-loss* ablation retains corruption-aware training but removes explicit trust supervision. Together they reveal a clean and scientifically useful separation.

First, corruption-aware training is the dominant source of task robustness. Removing it causes a pronounced drop in robust family accuracy and especially in window-level action accuracy, which falls from 0.9432 to 0.7533. Quality estimation also degrades substantially. Second, explicit trust supervision is primarily what makes the trust signal meaningful. When the trust loss is removed, the model maintains strong task performance, but trust RMSE becomes extremely poor under both clean and corrupted evaluation. This is an important

Representative controller phases



Object families used for evaluation

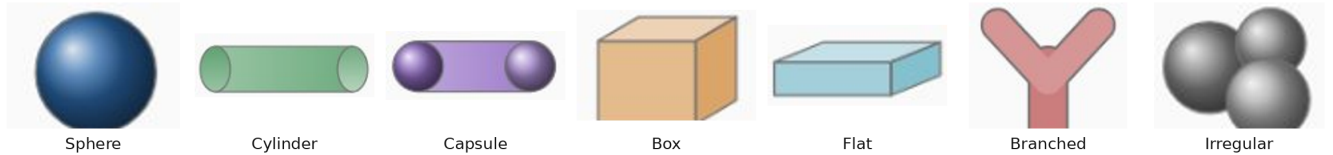


Fig. 2. Simulation protocol used for dataset generation. Top: representative controller phases from pack to pull. Bottom: the seven object families used for evaluation. The dataset intentionally spans both easy geometric primitives and harder nonconvex morphologies so that the learning problem reflects decision-level grasp diversity rather than a single object class.

TABLE IV
PLACEMENT-SPACE VALIDATION OF THE SELECTED 8-IMU SUBSET.
LOWER IS BETTER FOR BOTH METRICS.

Subset	Shape RMSE	Tip-angle error
D-optimal	0.0424	0.1038
Uniform	0.0546	0.1506
Hand-picked	0.0504	0.1091

result because it refines the paper’s core claim. The evidence does *not* suggest that trust supervision is the main driver of task robustness; rather, corruption-aware training produces robust task heads, while trust supervision makes the auxiliary reliability signal interpretable and informative.

This distinction strengthens the paper. It shows that the trust output is not a decorative side channel added after the fact. Instead, it measures something genuinely different from task accuracy: whether the current sensor set should be trusted, even when the model can still extract enough structure to make a correct decision.

C. Sparse Sensing, Actuation, and Label Granularity

A strong sparse-sensing paper must show not only that the final model works, but also that its sensing budget is justified. Figure 4 reports three design studies that support this claim: sensor-budget ablation, actuator-signal ablation, and family-label granularity. The message is consistent across all three.

The sensor-budget study shows that the D-optimal 8-IMU subset matches a 24-IMU layout within confidence intervals on

success, family accuracy, action accuracy, and quality RMSE. This is stronger than merely showing that an 8-IMU model performs well in isolation. It demonstrates that the sparse subset preserves essentially the same task-relevant information as the denser layout, which validates the placement strategy rather than just the downstream network.

The actuator-signal ablation leads to a similarly useful conclusion. Replacing the full simulated actuator state with displacement only leaves success, action, and quality nearly unchanged. This matters because displacement-only actuation is much closer to what the target hardware can measure reliably. The modest difference in family accuracy suggests that richer actuator signals may help finer object discrimination, but they are not necessary for the core grasp-assessment tasks.

Finally, the family-granularity study helps interpret what sparse proprioception can and cannot infer well. Exact-7 family accuracy is moderate at 0.500 ± 0.018 , but rises to 0.651 ± 0.040 for five coarser families and 0.672 ± 0.024 for three families. This indicates that the IMU-only signal captures broad mechanical response classes more readily than exact object identity. That is not a weakness for the main claim of the paper. On the contrary, it clarifies that sparse proprioception is especially strong for decision-level grasp assessment, while fine object taxonomy remains a more ambiguous secondary task.

VI. DISCUSSION

The empirical evidence supports four main conclusions.

First, sparse proprioception is sufficient for the core decision problem. The proposed model predicts grasp success

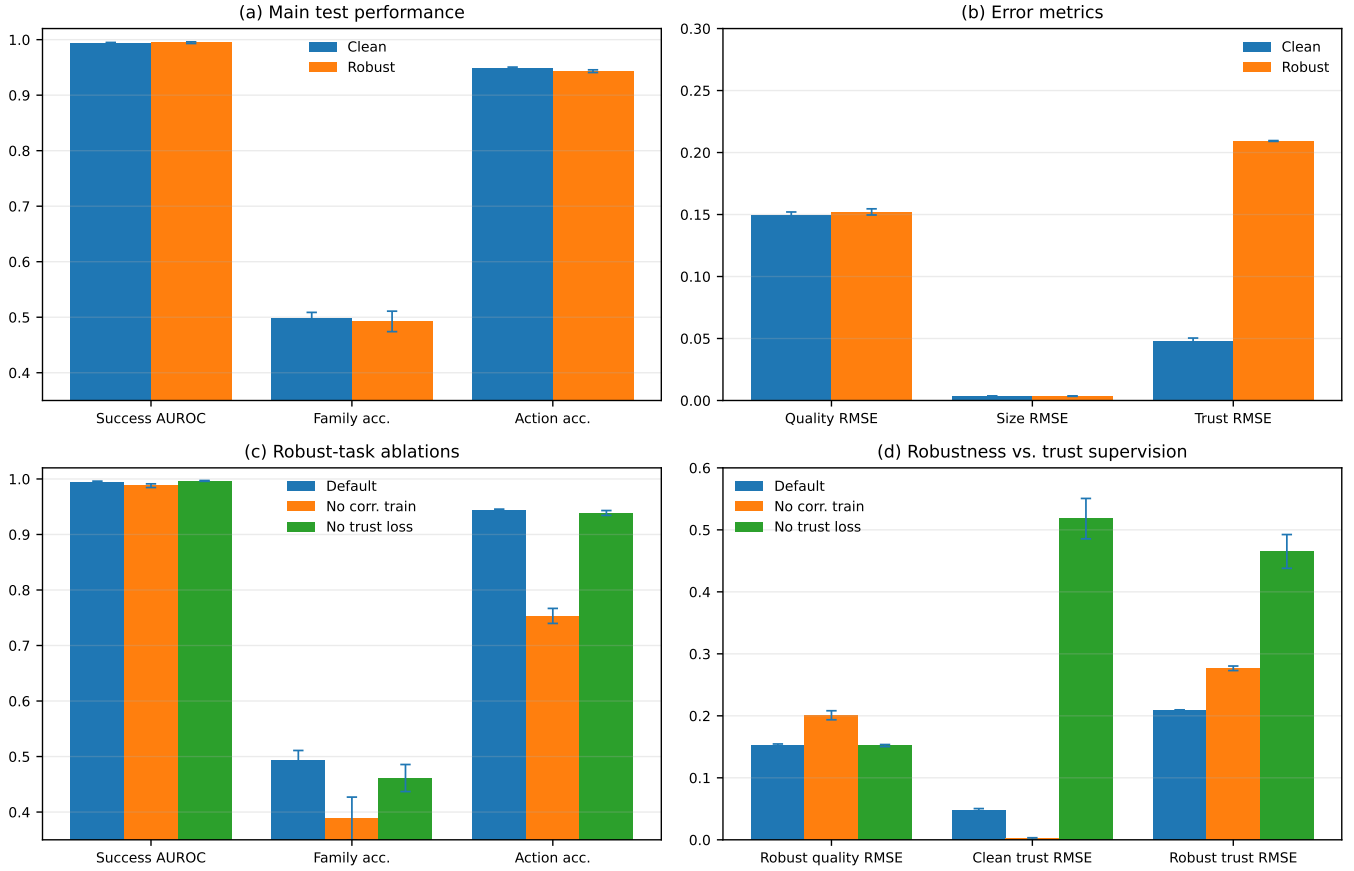


Fig. 3. Code-generated summary of the five-seed results. (a)–(b) Clean versus robust evaluation for the default model. (c)–(d) Robustness ablations. Removing corruption-aware training substantially damages robust family accuracy, action accuracy, and quality estimation, whereas removing trust loss mainly destroys trust estimation while leaving task performance comparatively intact.

and intervention labels reliably from short windows of body dynamics using only eight IMUs and displacement commands. The result is not merely that a compact model achieves high performance; it is that the sensing substrate itself contains enough information to support decision-level grasp assessment without cameras or tactile arrays.

Second, corruption-aware training is what makes the task heads robust. The no-corruption baseline confirms this directly: once online corruption is removed from training, robust action accuracy and family accuracy deteriorate sharply. This ablation is therefore central to the paper’s scientific credibility. It establishes a causal connection between the corruption-aware objective and the observed robustness, rather than leaving robustness as an incidental by-product of the architecture.

Third, trust supervision is essential, but for a narrower and more precise reason than is often implied in trustworthy-ML papers. The trust loss does not appear to be the main source of task robustness. Instead, it is what makes the trust output semantically meaningful. Once trust supervision is removed, the model still solves the main task well, but it loses the ability to report sensor reliability coher-

ently. This is precisely the kind of separation one wants in a trustworthy system: the task head remains useful, but the trust head explicitly signals when sensing reliability becomes hard to certify.

Fourth, the sparse design is empirically justified rather than heuristic. The D-optimal 8-IMU subset preserves performance relative to 24 IMUs and improves reconstruction-space metrics relative to uniform and hand-picked baselines. The displacement-only ablation further shows that the hardware-oriented sensing regime is not merely acceptable; it is already sufficient for the core outcome and action tasks.

These results also clarify the boundaries of the current contribution. The reported success AUROC is trial-level after aggregating overlapping windows, so it should be interpreted as strong *trial-level* grasp assessment rather than as proof of the earliest possible intervention point. Likewise, the action head imitates a rule-based oracle policy; high action accuracy therefore establishes strong policy imitation, not yet optimal closed-loop control. Finally, the trust targets arise from synthetic corruption models rather than measured hardware faults. None of these limitations invalidate the paper’s contribution. They simply define it accurately: this is a simulation-first,

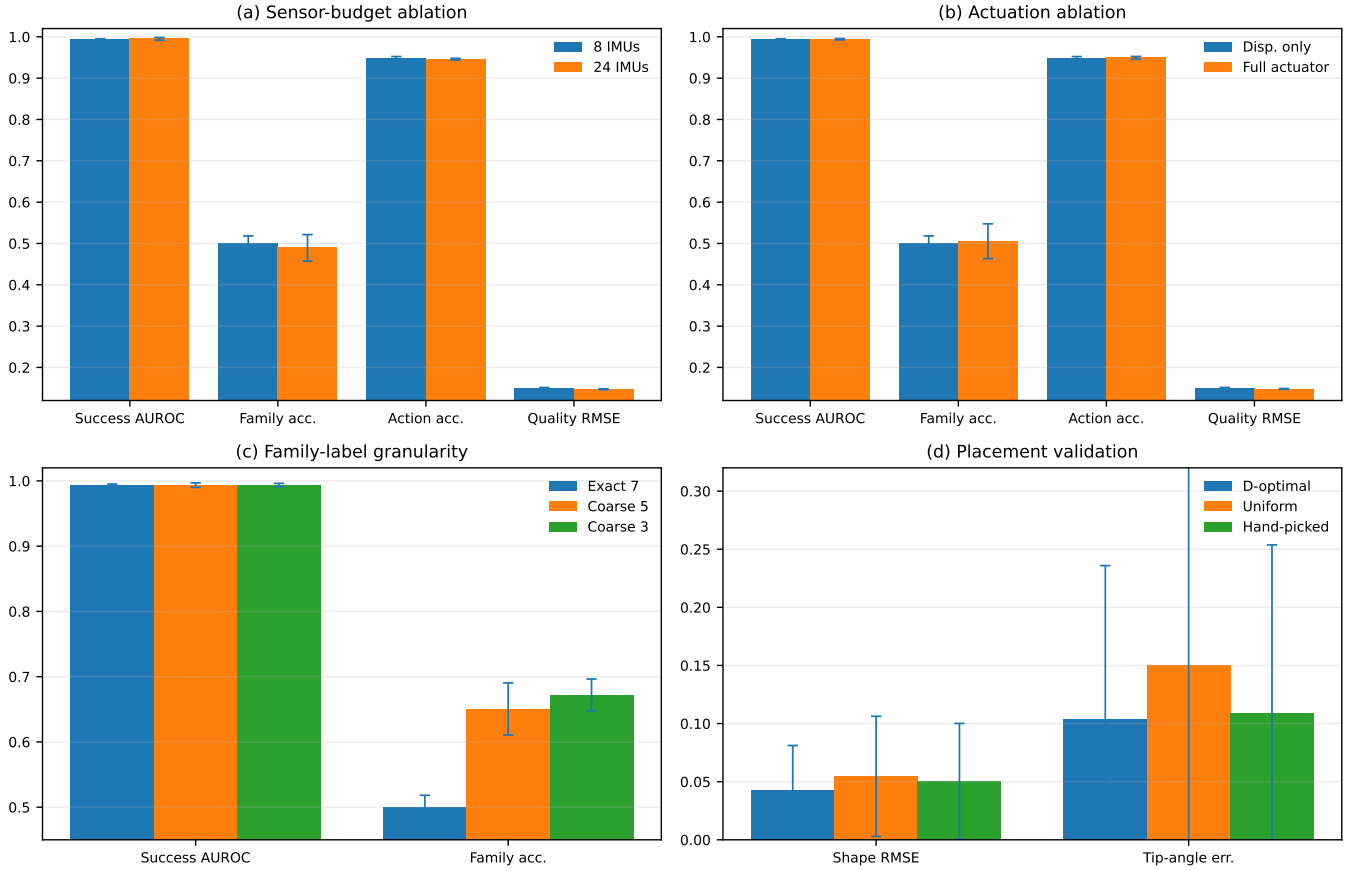


Fig. 4. Code-generated design ablations. (a) The D-optimal 8-IMU subset matches a 24-IMU layout within confidence intervals on the main clean-set metrics. (b) Restricting actuation to tendon displacement only leaves success and action essentially unchanged. (c) Family prediction becomes substantially easier as labels are mechanically coarsened from 7 classes to 5 and 3 classes. (d) The D-optimal subset improves reconstruction-space shape RMSE and tip-angle error relative to uniform and hand-picked 8-IMU baselines.

deployment-oriented machine-learning study whose strongest claims concern sparse sensing, robust task learning, and interpretable sensor reliability under corruption.

VII. CONCLUSION

We presented TrustGraspNet, a compact trust-aware framework for IMU-only grasp assessment in soft spiral robots under sparse sensing. The method combines D-optimal sensor placement, validity-aware temporal fusion, corruption-aware learning, and explicit per-IMU trust estimation in a deployment-oriented student model. The five-seed results show that eight carefully placed IMUs and displacement-only actuation are sufficient for strong trial-level grasp assessment and window-level intervention prediction, that corruption-aware training is the primary driver of robustness, and that explicit trust supervision is necessary to obtain meaningful reliability estimates. These findings position sparse proprioceptive grasp assessment as a credible contribution to both soft robotics and robust TinyML, while also offering a clear path toward future work on earlier intervention, held-out generalization, and hardware-backed trust calibration.

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